DESD Feb 2026 Embedded C Programming

Lab assignment-6 (Pointer)

Assignment 6a :

Q1. Write a C program for checking the endianness (little or big) of your current system.
Q2. Write a C program for Conversion of short integer from little endian to big endian(network order) and vice versa.
Q3. Write a C program for Conversion of integer from little endian to big endian(network order) and vice versa.
Q4. Given int a[5] = {10, 20, 30, 40, 50 }; int *p=a, *q = *(&a+1) - 1;
evaluate following expressions
*p++, *++p, (*p)++, ++(*p), ++*p, *(p++), *(++p)
*q--, *--q, --(*q), --*q,
(*q)--, *(q--), *(--q)
Q5. Write a C program to Convert from one type of pointer/address to other using void*
Q6. Write a C program to Test arithmetic operation on void pointers

Assignment 6b :

Q7. Given int arr[5]; int (*parr)[5]; parr=&arr;
Print sizeof(parr), sizeof(*parr), sizeof(**parr). Justify your answer.
Access array elements with suitable dereferencing of parr
Q8. Usage of assert macro before dereferencing any pointer.
Q9. Given int x;
Differentiate between const int * p; int const * p; int * const p = &x; const int * const p = &x;
Try *p=20, p++, (*p)++, p=&y in each case
Q10. Access 2D array using pointers int arr[3][4]; int (*p)[4]; p=arr;
sizeof(p), sizeof(*p), sizeof(**p) , values of p, p+1 Check equivalence of arr[i][j], (*(p+i))[j], *(*(p+i)+j)
Q11. Give an expression to print last element of array irrespective of length using pointer notation.
(You shouldn't consider length or size of array)
Q12. What is the significance of following pointer int (*q)[3][4];
What are the sizes of q, *q, **q, ***q
Write some code to test this with a 2D array.
